LAPORAN PRAKTIKUM
INSTALASI DAN KONFIGURASI WEB SERVER

OLEH:

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D3 TEKNOLOGI KOMPUTER

INSTITUT TEKNOLOGI DEL

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INSTALASI WEB SERVER

1. Install Apache2 sebagai web server utama dengan perintah berikut

```
aisasilvana@aisasilvana:~$ sudo apt install apache2 -y
[sudo] password for aisasilvana:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.11).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
aisasilvana@aisasilvana:~$
```

2. Setelah instalasi selesai, cek status service Apache

```
aisasilvana@aisasilvana:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-04-23 00:38:27 UTC; 8min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 1378 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1416 (apache2)
      Tasks: 55 (limit: 4544)
     Memory: 7.4M (peak: 7.8M)
        CPU: 156ms
     CGroup: /system.slice/apache2.service
             ├─1416 /usr/sbin/apache2 -k start
             ├─1420 /usr/sbin/apache2 -k start
             └─1421 /usr/sbin/apache2 -k start

Apr 23 00:38:27 aisasilvana systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 23 00:38:27 aisasilvana apachectl[1403]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the >
Apr 23 00:38:27 aisasilvana systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
```

3. Setelah instalasi Apache selesai dan status service berhasil dicek, langkah berikutnya adalah memastikan server terhubung ke internet

```
aisasilvana@aisasilvana:~$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=65.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=109 ms
^C
--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 65.577/87.418/109.260/21.841 ms
aisasilvana@aisasilvana:~$ sudo apt install mysql-server php php-mysql php-curl php-gd php-mbstring php-xml php-xmlrpc php-soap php-intl php-zip libapache2-mod-
php -y
[sudo] password for aisasilvana:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2
  libprotobuf-lite32t64 libtimedate-perl liburi-perl libxmlrpc-epi0t64 libzip4t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server-8.0 mysql-server-core-8.0 php-common php8.3 php8.3-cli php8.3-common php8.3-curl php8.3-gd php8.3-intl
  php8.3-mbstring php8.3-mysql php8.3-opcache php8.3-readline php8.3-soap php8.3-xml php8.3-xmlrpc php8.3-zip
Suggested packages:
  php-pear libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl
  liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl libxmlrpc-epi0t64 libzip4t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils
  mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0 php php-common php-curl php-gd php-intl php-mbstring
  php-mysql php-soap php-xml php-xmlrpc php-zip php8.3 php8.3-cli php8.3-common php8.3-curl php8.3-gd php8.3-intl php8.3-mbstring php8.3-mysql php8.3-opcache
  php8.3-readline php8.3-soap php8.3-xml php8.3-xmlrpc php8.3-zip
0 upgraded, 57 newly installed, 0 to remove and 2 not upgraded.
Need to get 35,8 MB of archives.
After this operation, 269 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6.746 B]
Get:2 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [2.740 kB]
Get:3 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [22,5 kB]
Get:4 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7.982 B]
Get:5 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:6 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.3 [238 kB]
Get:7 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server-core-8.0 amd64 8.0.45-0ubuntu0.24.04.1 [17,6 MB]
19% [7 mysql-server-core-8.0 4.160 kB/17,6 MB 24%]                                                                                    886 kB/s 32s
```

4. Setelah koneksi internet dipastikan berjalan dengan baik, lanjutkan dengan instalasi MySQL Server menggunakan perintah berikut

```
aisasilvana@aisasilvana:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> _
```

5. Kemudian buat Database WordPress jalankan perintah berikut

```
CREATE DATABASE wordpress;

CREATE USER 'wpuser'@'localhost' IDENTIFIED BY 'password123';

GRANT ALL PRIVILEGES ON wordpress.* TO 'wpuser'@'localhost';

FLUSH PRIVILEGES;

EXIT;
```

```
mysql> CREATE DATABASE wordpress;
CREATE Query OK, 1 row affected (0,01 sec)

mysql> CREATE USER 'wpuser'@'localhost' IDENTIFIED BY 'password123';
GRANT ALL PRIVILQuery OK, 0 rows affected (0,03 sec)E

mysql> GRANT ALL PRIVILEGES ON wordpress.* TO 'wpuser'@'localhost';
FLUSQuery OK, 0 rows affected (0,00 sec)

mysql> FLUSH PRIVILEGES;
EXQuery OK, 0 rows affected (0,01 sec)

Imysql> EXIT;
Bye
aisasilvana@aisasilvana:~$ _
```

6. Setelah instalasi Apache, PHP, dan MySQL selesai, langkah berikutnya adalah mengirim file website dari komputer Windows ke Ubuntu Server menggunakan perintah scp.

```
PS C:\Users\Acer> scp "C:\xampp2\htdocs\wordpress.zip" aisasilvana@192.168.2
30.130:/home/aisasilvana/
aisasilvana@192.168.230.130's password:
wordpress.zip                                 100%  234MB  68.3MB/s   00:03
PS C:\Users\Acer>
```

7. Setelah folder website berhasil dikirim, lanjutkan dengan mengirim file database Perintah ini digunakan untuk mengirim file database SQL agar dapat diimport ke MySQL server Ubuntu.Jika berhasil, file smansib.sql akan tersimpan di server

```
PS C:\Users\Acer> scp "D:\smansasib.sql" aisasilvana@192.168.230.130:/home/a
isasilvana/
aisasilvana@192.168.230.130's password:
smansasib.sql                                 100%   36MB  55.1MB/s   00:00
PS C:\Users\Acer>
```

8. Setelah file website dan database berhasil dikirim menggunakan scp, Langkah berikutnya adalah masuk ke direktori home user untuk memastikan file sudah berhasil masuk.

```
aisasilvana@aisasilvana:~$ cd /home/aisasilvana
aisasilvana@aisasilvana:~$ ls
smansasib.sql  wordpress.zip
aisasilvana@aisasilvana:~$
```

9. Setelah file website berhasil dikirim ke direktori /home/aisasilvana, jika file masih dalam bentuk .zip, maka langkah berikutnya adalah mengekstraknya menggunakan perintah unzip.

```
aisasilvana@aisasilvana:~$ sudo unzip /home/aisasilvana/wordpress.zip -d /var/www/html/
```

10. Selanjutnya pindahkan folder website ke direktori Apache dan Sebelum menggunakan website smansib, folder WordPress yang sebelumnya ada di direktori Apache perlu dihapus agar tidak terjadi konflik dengan website yang baru. Lalu Setelah folder website smansib dipindahkan ke direktori Apache dan folder WordPress dihapus, langkah berikutnya adalah mengatur hak akses file agar Apache dapat membaca dan menjalankan website dengan baik. Maka jalankan perintah berikut

```
aisasilvana@aisasilvana:~$ sudo mv /var/www/html/wordpress/* /var/www/html/
aisasilvana@aisasilvana:~$ sudo rm -rf /var/www/html/wordpress
aisasilvana@aisasilvana:~$ sudo chown -R www-data:www-data /var/www/html/
aisasilvana@aisasilvana:~$ sudo chmod -R 755 /var/www/html/
aisasilvana@aisasilvana:~$
```

```
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/embed-404.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/embed-content.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/embed.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/footer-embed.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/footer.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/header-embed.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/header.php
  inflating: /var/www/html/wordpress/wp-includes/theme-compat/sidebar.php
  inflating: /var/www/html/wordpress/wp-includes/theme-i18n.json
  inflating: /var/www/html/wordpress/wp-includes/theme-previews.php
  inflating: /var/www/html/wordpress/wp-includes/theme-templates.php
  inflating: /var/www/html/wordpress/wp-includes/theme.json
  inflating: /var/www/html/wordpress/wp-includes/theme.php
  inflating: /var/www/html/wordpress/wp-includes/update.php
  inflating: /var/www/html/wordpress/wp-includes/user.php
  inflating: /var/www/html/wordpress/wp-includes/vars.php
  inflating: /var/www/html/wordpress/wp-includes/version.php
   creating: /var/www/html/wordpress/wp-includes/widgets/
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-nav-menu-widget.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-archives.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-block.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-calendar.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-categories.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-custom-html.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-links.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-media-audio.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-media-gallery.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-media-image.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-media-video.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-media.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-meta.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-pages.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-recent-comments.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-recent-posts.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-rss.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-search.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-tag-cloud.php
  inflating: /var/www/html/wordpress/wp-includes/widgets/class-wp-widget-text.php
  inflating: /var/www/html/wordpress/wp-includes/widgets.php
  inflating: /var/www/html/wordpress/wp-includes/wp-db.php
  inflating: /var/www/html/wordpress/wp-includes/wp-diff.php
  inflating: /var/www/html/wordpress/wp-links-opml.php
  inflating: /var/www/html/wordpress/wp-load.php
  inflating: /var/www/html/wordpress/wp-login.php
  inflating: /var/www/html/wordpress/wp-mail.php
  inflating: /var/www/html/wordpress/wp-settings.php
  inflating: /var/www/html/wordpress/wp-signup.php
  inflating: /var/www/html/wordpress/wp-trackback.php
  inflating: /var/www/html/wordpress/xmlrpc.php
aisasilvana@aisasilvana:~$
```

11. Setelah permission folder website selesai diatur, langkah berikutnya adalah melakukan konfigurasi file database pada website agar dapat terhubung dengan MySQL Server. Kemudian edit file konfigurasi:

```
  GNU nano 7.2                                                    /var/www/html/wp-config.php *
<?php
/**
 * The base configuration for WordPress
 *
 * The wp-config.php creation script uses this file during the installation.
 * You don't have to use the website, you can copy this file to "wp-config.php"
 * and fill in the values.
 *
 * This file contains the following configurations:
 *
 * * Database settings
 * * Secret keys
 * * Database table prefix
 * * ABSPATH
 *
 * @link https://developer.wordpress.org/advanced-administration/wordpress/wp-config/
 *
 * @package WordPress
 */

// ** Database settings - You can get this info from your web host ** //
/** The name of the database for WordPress */
define( 'DB_NAME', 'wordpress' );

/** Database username */
define( 'DB_USER', 'wpuser' );

/** Database password */
define( 'DB_PASSWORD', 'password123' );

/** Database hostname */
define( 'DB_HOST', 'localhost' );

/** Database charset to use in creating database tables. */
define( 'DB_CHARSET', 'utf8mb4' );

/** The database collate type. Don't change this if in doubt. */
define( 'DB_COLLATE', '' );
```

12. Setelah file konfigurasi website selesai diedit, langkah berikutnya adalah masuk ke MySQL Server untuk membuat database dan memastikan koneksi database sesuai dengan konfigurasi pada wp-config.php.

```
aisasilvana@aisasilvana:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE smansasib;

GRANTQuery OK, 1 row affected (0,01 sec)

mysql>
mysql> GRANT ALL PRIVILEGES ON smansasib.* TO 'root'@'localhost';

FLUSQuery OK, 0 rows affected (0,01 sec)

mysql>
mysql> FLUSH PRIVILEGES;

EXQuery OK, 0 rows affected (0,00 sec)

mysql> I
mysql> EXIT;
```
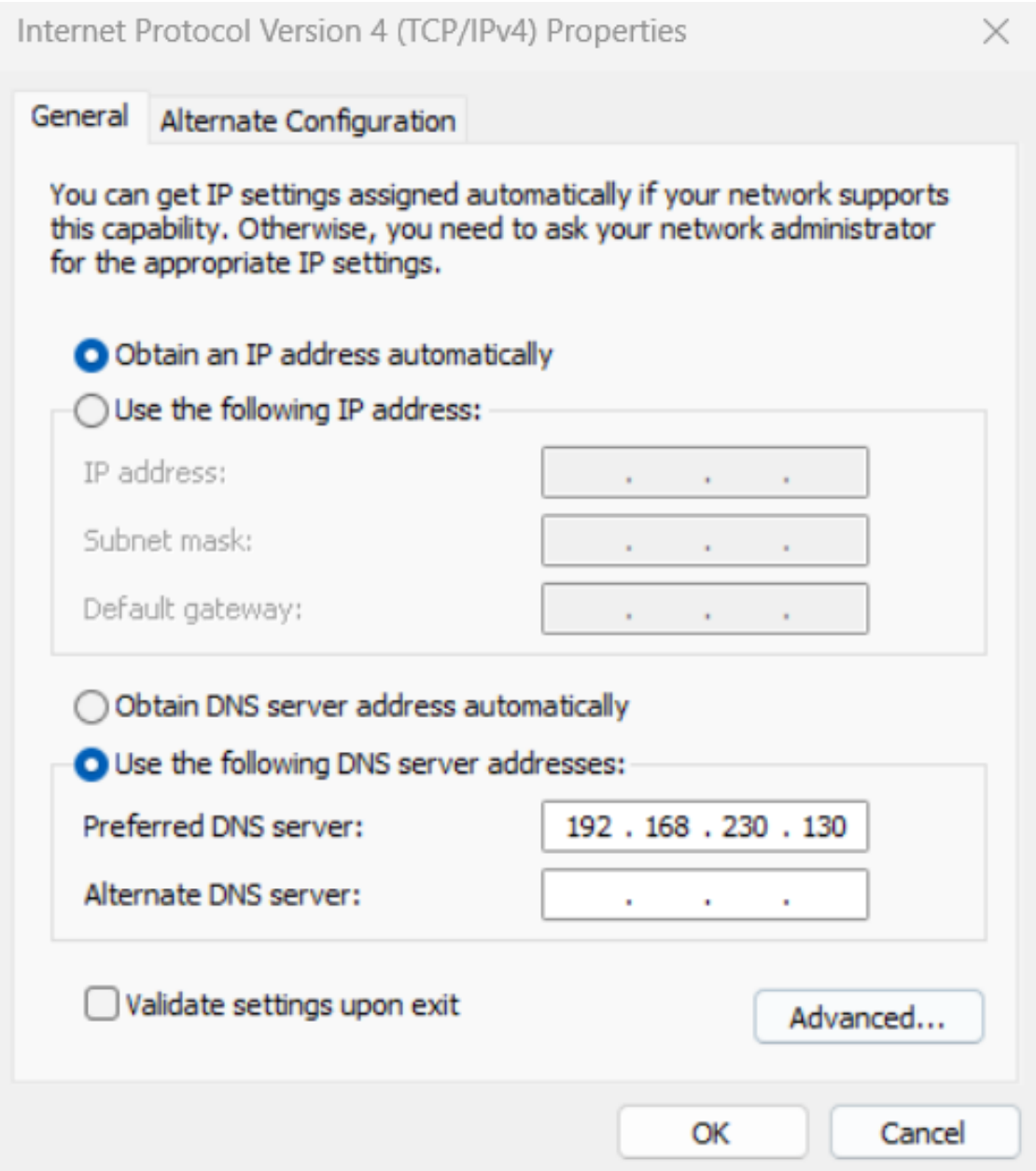
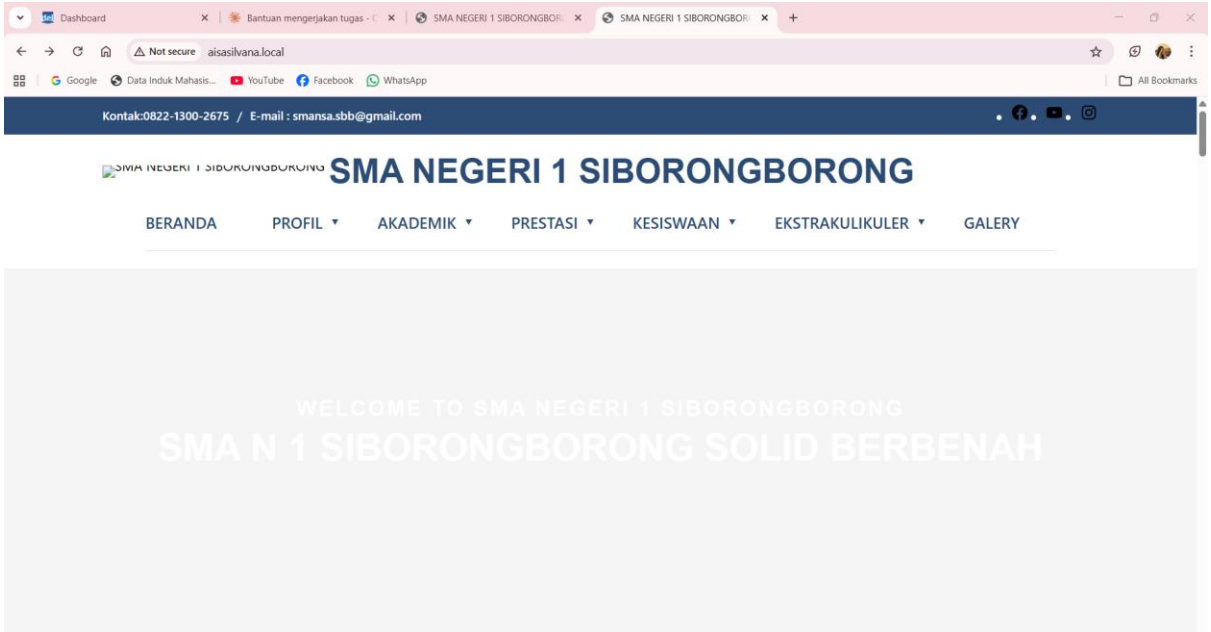

13. Setelah konfigurasi database selesai, langkah berikutnya adalah me-restart service agar seluruh perubahan dapat diterapkan dengan baik. Gunakan perintah berikut:

```
aisasilvana@aisasilvana:~$ sudo systemctl restart apache2
aisasilvana@aisasilvana:~$ _
```

14. Setelah DHCP, DNS, dan Web Server selesai dikonfigurasi, komputer client perlu diatur pada bagian TCP/IPv4 agar dapat terhubung ke Ubuntu Server.



15. Setelah itu coba jalankan di google maupun browser anda



IP CLIENT

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:24:cc:b7 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.100.10/24 brd 192.168.100.255 scope global dynamic noprefixrout
e ens33
       valid_lft 507sec preferred_lft 507sec
    inet6 fe80::20c:29ff:fe24:ccb7/64 scope link
       valid_lft forever preferred_lft forever
```

IP UBUNTU SERVER

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ff:f9:78 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.230.130/24 metric 100 brd 192.168.230.255 scope global dynamic ens33
       valid_lft 1653sec preferred_lft 1653sec
    inet6 fe80::20c:29ff:feff:f978/64 scope link
       valid_lft forever preferred_lft forever
```